

Corrected Convergence Guarantees for Finite Time-Inconsistent Equilibrium Transport

Technical note

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Abstract

Bayraktar and Han ask for convergence guarantees for numerical algorithms for equilibrium transport with time-inconsistent costs. This note gives a corrected finite-dimensional result and identifies the precise limitation of a previously proposed argument. The broad open problem is *not* solved; the results constitute partial progress. In a fully finite model, continuity and uniqueness of every one-step best response imply convergence of inexact backward sweeps and of a parallel Jacobi best-response iteration, even without a contraction condition. The limit is the unique policy that is optimal at every marginally feasible history. Such a policy always induces an equilibrium in the almost-sure sense of Bayraktar–Han. However, strong convexity alone does not imply uniqueness for their equilibrium notion: an explicit two-period binary counterexample exhibits two distinct equilibrium transport plans despite smooth strongly convex one-step objectives. We recover equivalence and uniqueness under an interior-best-response condition. This condition holds for a natural finite class of f -divergence regularized problems when the generator is strictly convex and satisfies $f'(0+) = -\infty$, including the squared Hellinger and Jensen–Shannon generators highlighted as time-inconsistent examples by Bayraktar and Han. A separate quantitative result covers smooth strongly convex local objectives and projected-gradient subsolvers.

Status. This paper gives partial progress, not a solution of the general open problem. It proves convergence for fully finite instances under local uniqueness and continuity, and proves convergence to the unique Bayraktar–Han equilibrium under an additional interiority condition. It does not address general semi-discrete or continuous models, discretization limits, or arbitrary nonconvex and nonunique local problems.

1 Introduction

Bayraktar and Han introduce equilibrium bicausal transport for time-inconsistent costs and state that convergence of numerical algorithms remains open; see Remark 2.1 of [1] and the problem record [2]. Their finite discrete analysis already supplies an exact backward extended dynamic-programming recursion and notes that multiple equilibrium transports can occur. Consequently, an exact backward construction in the finite model is not by itself a resolution of the open problem. A useful numerical result should at least explain what happens when the local optimization problems are solved only approximately and should state carefully which equilibrium notion is computed.

The main correction in this note concerns off-path histories. Bayraktar and Han impose one-step optimality at time t only for histories that occur almost surely under the candidate transport plan. Requiring optimality at *every* marginally feasible history is stronger. The two conditions coincide when the equilibrium policy has full support, but they are not equivalent in general, even with

full-support marginals and smooth strongly convex local objectives. Section 3 gives a minimal two-period binary counterexample.

After separating these notions, the triangular structure becomes useful and rigorous. We prove:

- (i) a qualitative convergence theorem for inexact backward sweeps and a synchronous Jacobi best-response iteration under only continuity and unique local minimizers;
- (ii) quantitative error propagation and a correct projected-gradient complexity bound under smooth strong convexity;
- (iii) equivalence with the Bayraktar–Han equilibrium notion under an interior-best-response condition; and
- (iv) verification of that interiority condition for a finite class of barrier-type f -divergence regularizers.

The exact backward sweep is essentially the discrete extended dynamic program already present in [1]. The additional content is robustness to inexact local solves, convergence of a parallel delayed update, the counterexample to the unqualified uniqueness claim, and conditions that restore uniqueness for the authors’ equilibrium notion.

2 Finite policy model and two equilibrium notions

Fix a horizon $T \geq 1$. For $t = 1, \dots, T$, let X_t and Y_t be finite sets, and let μ and ν be probability laws on $X_{1:T}$ and $Y_{1:T}$. We discard marginal histories of zero probability. Thus

$$\mathcal{H}_t := \text{supp}(\mu_{1:t}) \times \text{supp}(\nu_{1:t}), \quad t = 1, \dots, T-1,$$

and $\mathcal{H}_0 = \{\emptyset\}$. If $h = (x_{1:t}, y_{1:t}) \in \mathcal{H}_t$, define

$$m_h^X(x) = \mu(x_{t+1} = x \mid x_{1:t}), \quad m_h^Y(y) = \nu(y_{t+1} = y \mid y_{1:t}),$$

and the local transportation polytope

$$\mathcal{K}_h := \left\{ \gamma \geq 0 : \sum_y \gamma(x, y) = m_h^X(x), \quad \sum_x \gamma(x, y) = m_h^Y(y) \right\}.$$

Zero rows and columns may be deleted. With this convention, the relative interior $\text{ri } \mathcal{K}_h$ consists exactly of matrices that are strictly positive on every active cell

$$S_h := \text{supp}(m_h^X) \times \text{supp}(m_h^Y).$$

A *policy* is a collection $a = (a_h)_{h \in \cup_t \mathcal{H}_t}$ with $a_h \in \mathcal{K}_h$. Its policy space

$$\mathcal{A} := \prod_{t=0}^{T-1} \prod_{h \in \mathcal{H}_t} \mathcal{K}_h$$

is compact. The induced bicausal coupling is

$$\Pi(a)(x_{1:T}, y_{1:T}) = \prod_{t=0}^{T-1} a_{(x_{1:t}, y_{1:t})}(x_{t+1}, y_{t+1}). \quad (1)$$

For a history $h \in \mathcal{H}_t$, let $\text{Desc}(h)$ be the set of its strict descendants, and write $a_{\text{Desc}(h)}$ for the corresponding subpolicy. Let

$$\Phi_h(\gamma, b), \quad \gamma \in \mathcal{K}_h, \quad b \in \mathcal{A}_{\text{Desc}(h)},$$

be the objective of the time- t planner at history h when the current one-step kernel is γ and all later kernels are b . This is the finite-policy version of the one-step perturbation objective in Definition 2.2 of [1].

For $a \in \mathcal{A}$, let $p_a(h)$ denote the probability of reaching h under $\Pi(a)$; in particular, $p_a(\emptyset) = 1$.

Definition 1 (Bayraktar–Han equilibrium policy). A globally specified policy $a \in \mathcal{A}$ is a *BH equilibrium policy* if, for every t and every $h \in \mathcal{H}_t$ with $p_a(h) > 0$,

$$a_h \in \arg \min_{\gamma \in \mathcal{K}_h} \Phi_h(\gamma, a_{\text{Desc}(h)}). \quad (2)$$

Its induced measure $\Pi(a)$ is a Bayraktar–Han equilibrium transport plan. The qualification $p_a(h) > 0$ is the finite form of their almost-sure condition.

Because a one-step deviation can make an off-path history reachable, Definition 1 is understood relative to the globally specified versions of all future kernels. This is also how the perturbed plan in [1, Definition 2.2] is evaluated.

Definition 2 (All-histories equilibrium). A policy $a \in \mathcal{A}$ is an *all-histories equilibrium* if (2) holds for every $h \in \cup_t \mathcal{H}_t$, whether or not $p_a(h) > 0$.

Proposition 3. *Every all-histories equilibrium is a BH equilibrium policy.*

Proof. The all-histories condition contains the BH condition as a subset of its constraints. $\square$

The converse is false. This is the source of the incorrect uniqueness claim in the original note.

3 A two-period counterexample

Proposition 4 (Strong convexity does not imply BH uniqueness). *There is a two-period finite problem with full-support marginals such that every local objective is a smooth strongly convex function of the current coupling and depends Lipschitz-continuously on future kernels, yet there are two distinct BH equilibrium transport plans. The unique all-histories equilibrium is only one of them.*

Proof. Let $T = 2$ and

$$X_1 = Y_1 = X_2 = Y_2 = \{0, 1\}.$$

Let both marginal processes be independent fair Bernoulli processes. Every coupling of two fair Bernoulli variables has the form

$$K(r) = \frac{1}{2} \begin{pmatrix} r & 1-r \\ 1-r & r \end{pmatrix}, \quad 0 \leq r \leq 1. \quad (3)$$

The parameter r is exactly the probability that the two variables are equal.

Write $r_\emptyset$ for the first-stage parameter and r_{ij} for the second-stage parameter after history $(X_1, Y_1) = (i, j)$. Define the time-1 cost by

$$J_1(i, j; \pi) = \mathbb{P}_\pi(X_2 = Y_2 \mid X_1 = i, Y_1 = j)^2. \quad (4)$$

Hence the local time-1 objective is r_{ij}^2 , whose unique minimizer is $r_{ij} = 0$.

Define the time-0 cost by

$$J_0(\pi) = \frac{1}{2}(\mathbb{P}_\pi(X_1 = Y_1) - 1)^2 + \mathbb{E}_\pi[\mathbf{1}_{\{X_1=Y_1\}}\mathbf{1}_{\{X_2=Y_2\}}]. \quad (5)$$

If the first-stage parameter is changed to r while the future parameters are fixed, then

$$\Phi_\emptyset(r, (r_{ij})) = \frac{1}{2}(r - 1)^2 + \frac{r}{2}(r_{00} + r_{11}). \quad (6)$$

Indeed, the two diagonal first-stage histories each have probability $r/2$. The objective in (6) has second derivative 1, and the terminal objectives have second derivative 2. Their gradients are Lipschitz in all variables. Since $r \mapsto K(r)$ is an affine bijection onto the one-dimensional transportation polytope, these scalar curvature bounds are exactly smooth strong-convexity bounds in reduced affine coordinates.

The all-histories backward equations first force

$$r_{00} = r_{01} = r_{10} = r_{11} = 0.$$

Equation (6) then has the unique minimizer $r_\emptyset = 1$. Thus the unique all-histories equilibrium is

$$r_\emptyset = 1, \quad r_{ij} = 0 \quad \text{for all } i, j. \quad (7)$$

Now consider the different policy

$$r_\emptyset = 0, \quad r_{01} = r_{10} = 0, \quad r_{00} = r_{11} = 1. \quad (8)$$

Under this policy, only the histories $(0, 1)$ and $(1, 0)$ occur at time 1. At both reached histories, the chosen parameter 0 is the unique minimizer of (4). At time 0, a deviation to $r \in [0, 1]$ is evaluated using the fixed off-path continuation values $r_{00} = r_{11} = 1$. Therefore

$$\Phi_\emptyset(r, (r_{ij})) = \frac{1}{2}(r - 1)^2 + r = \frac{1}{2}r^2 + \frac{1}{2},$$

whose unique minimizer is $r = 0$. Hence (8) is a BH equilibrium policy.

The plans induced by (7) and (8) are distinct: under the first, $\mathbb{P}(X_1 = Y_1) = 1$, while under the second this probability is 0. This proves the claim. $\square$

Remark 5. The counterexample shows that full support of μ and ν is insufficient: a coupling of full-support marginals can assign zero probability to joint histories. It also shows that local smoothness, strong convexity, and Lipschitz dependence on future kernels do not by themselves imply uniqueness of a BH equilibrium plan.

4 Inexact convergence to the all-histories equilibrium

The basic convergence theorem needs neither differentiability nor a contraction.

Assumption 6 (Continuity and unique local minimizers). For every history h :

- (a) Φ_h is continuous on the compact set $\mathcal{K}_h \times \mathcal{A}_{\text{Desc}(h)}$;

(b) for every $b \in \mathcal{A}_{\text{Desc}(h)}$, the problem

$$\min_{\gamma \in \mathcal{K}_h} \Phi_h(\gamma, b)$$

has a unique minimizer.

Define

$$v_h(b) := \min_{\gamma \in \mathcal{K}_h} \Phi_h(\gamma, b), \quad \text{BR}_h(b) := \arg \min_{\gamma \in \mathcal{K}_h} \Phi_h(\gamma, b).$$

Lemma 7 (Stability of approximate minimizers). *Under Assumption 6, v_h and BR_h are continuous. More generally, if $b_n \rightarrow b$, $\delta_n \rightarrow 0$, and $u_n \in \mathcal{K}_h$ satisfies*

$$\Phi_h(u_n, b_n) \leq v_h(b_n) + \delta_n, \tag{9}$$

then $u_n \rightarrow \text{BR}_h(b)$.

Proof. Continuity of v_h follows from the compact-set form of Berge's maximum theorem; it can also be proved directly by uniform continuity. To prove the last assertion, take an arbitrary convergent subsequence $u_{n_k} \rightarrow u$, which exists by compactness of $\mathcal{K}_h$. Passing to the limit in (9) gives

$$\Phi_h(u, b) \leq v_h(b).$$

Thus $u = \text{BR}_h(b)$ by uniqueness. Every convergent subsequence has the same limit, so the full sequence converges. Taking $u_n = \text{BR}_h(b_n)$ and $\delta_n = 0$ proves continuity of BR_h . $\square$

4.1 Inexact backward sweeps

For each accuracy index N , construct a policy $\hat{a}^N$ backward in time. Once all strict descendants of h have been computed, choose $\hat{a}_h^N \in \mathcal{K}_h$ satisfying

$$\Phi_h(\hat{a}_h^N, \hat{a}_{\text{Desc}(h)}^N) \leq v_h(\hat{a}_{\text{Desc}(h)}^N) + \delta_{h,N}, \tag{10}$$

where

$$\bar{\delta}_N := \max_h \delta_{h,N} \rightarrow 0.$$

Theorem 8 (Inexact backward-sweep convergence). *Under Assumption 6, there is a unique all-histories equilibrium $a^\sharp$. Every sequence of inexact backward sweeps satisfying (10) and $\bar{\delta}_N \rightarrow 0$ obeys*

$$\hat{a}_h^N \rightarrow a_h^\sharp \quad \text{for every history } h.$$

If all local problems are solved exactly, one backward sweep returns $a^\sharp$.

Proof. At terminal histories $h \in \mathcal{H}_{T-1}$ there are no descendants. Define

$$a_h^\sharp := \text{BR}_h(\emptyset).$$

Lemma 7 applied to (10) gives $\hat{a}_h^N \rightarrow a_h^\sharp$. Suppose recursively that $a_g^\sharp$ has been defined and $\hat{a}_g^N \rightarrow a_g^\sharp$ for every strict descendant g of a history $h \in \mathcal{H}_t$. Set

$$a_h^\sharp := \text{BR}_h(a_{\text{Desc}(h)}^\sharp).$$

Then $\hat{a}_{\text{Desc}(h)}^N \rightarrow a_{\text{Desc}(h)}^\sharp$, and Lemma 7 again yields $\hat{a}_h^N \rightarrow a_h^\sharp$. Backward induction proves convergence at every history. The same recursion proves existence and uniqueness of $a^\sharp$. When every $\delta_{h,N} = 0$, the backward construction agrees with this recursion exactly. $\square$

The exact part of Theorem 8 is the finite extended dynamic-programming construction already implicit in [1, Section 2.3]. The convergence statement for vanishing numerical errors is the relevant robustness conclusion.

4.2 A parallel Jacobi iteration

A delayed update permits all histories to be processed in parallel once the previous policy is available. Starting from arbitrary $a^0 \in \mathcal{A}$, choose a^{n+1} so that

$$\Phi_h(a_h^{n+1}, a_{\text{Desc}(h)}^n) \leq v_h(a_{\text{Desc}(h)}^n) + \delta_{h,n}, \quad \bar{\delta}_n := \max_h \delta_{h,n} \rightarrow 0. \quad (11)$$

Theorem 9 (Parallel inexact Jacobi convergence). *Under Assumption 6, every sequence satisfying (11) converges coordinatewise to the unique all-histories equilibrium $a^\sharp$, from every initialization. If $\delta_{h,n} = 0$ for all h, n , then $a^n = a^\sharp$ for every $n \geq T$.*

Proof. For terminal histories, (11) and Lemma 7 imply $a_h^{n+1} \rightarrow a_h^\sharp$. Suppose all stages strictly later than t converge. Then $a_{\text{Desc}(h)}^n \rightarrow a_{\text{Desc}(h)}^\sharp$ for each $h \in \mathcal{H}_t$, so Lemma 7 applied to (11) gives $a_h^{n+1} \rightarrow a_h^\sharp$. Backward induction proves convergence.

For exact updates, all terminal kernels are correct after one iteration. If all stages later than t are correct at iteration n , then stage t is correct at iteration $n + 1$. Therefore all stages are correct after at most T iterations. $\square$

Lemma 10 (Policy convergence implies plan convergence). *For any two policies $a, a' \in \mathcal{A}$,*

$$\|\Pi(a) - \Pi(a')\|_1 \leq \sum_{t=0}^{T-1} \max_{h \in \mathcal{H}_t} \|a_h - a'_h\|_1. \quad (12)$$

Consequently, coordinatewise policy convergence implies convergence in total variation, where $\|P - Q\|_{\text{TV}} := \frac{1}{2} \|P - Q\|_1$.

Proof. For a fixed path, telescope the difference of the two products in (1), using a in factors before time t and a' in factors after time t . After taking absolute values and summing over all paths, the contribution of time t is

$$\sum_{h \in \mathcal{H}_t} p_a(h) \|a_h - a'_h\|_1 \leq \max_{h \in \mathcal{H}_t} \|a_h - a'_h\|_1,$$

because the prefix probabilities sum to one. Summing over t gives (12). $\square$

Corollary 11. *The plans produced by Theorems 8 and 9 converge in total variation to $\Pi(a^\sharp)$.*

5 When the computed policy is the unique BH equilibrium

The missing condition in the unqualified uniqueness claim is positivity of local best responses.

Assumption 12 (Interior best responses). *For every history h and every descendant policy b ,*

$$\text{BR}_h(b) \in \text{ri } \mathcal{K}_h.$$

Equivalently, the best response assigns positive mass to every active one-step cell.

Theorem 13 (Equivalence and BH uniqueness under interiority). *Suppose Assumption 6 holds.*

- (a) *The all-histories equilibrium $a^\sharp$ is a BH equilibrium policy.*
- (b) *If Assumption 12 also holds, then every BH equilibrium policy is an all-histories equilibrium. Hence $a^\sharp$ is the unique BH equilibrium policy and $\Pi(a^\sharp)$ is the unique BH equilibrium transport plan.*

Thus, under Assumption 12, Theorems 8 and 9 converge to the unique equilibrium in the exact sense of Bayraktar and Han.

Proof. Part (a) is Proposition 3. For part (b), let a be a BH equilibrium. The root is reached, so $a_\emptyset = \text{BR}_\emptyset(a_{\text{Desc}(\emptyset)}) \in \text{ri } \mathcal{K}_\emptyset$. Hence every active time-1 joint history has positive probability. At each such history, the BH condition applies and Assumption 12 makes every active child have positive conditional probability. Inductively, every history in every $\mathcal{H}_t$ is reached. The BH optimality condition therefore holds at every history, so a is an all-histories equilibrium. Uniqueness follows from Theorem 8. $\square$

6 Quantitative bounds in the smooth strongly convex case

This section recovers the valid quantitative part of the original argument. For each h , identify the affine hull of $\mathcal{K}_h$ with a Euclidean space using orthonormal reduced coordinates. All gradients and norms below refer to those coordinates.

Assumption 14 (Strong convexity and future-kernel sensitivity). For every history h , there are constants $\sigma_h > 0$ and $M_h < \infty$ such that, for every descendant policy b :

- (a) $\gamma \mapsto \Phi_h(\gamma, b)$ is differentiable and σ_h -strongly convex on $\mathcal{K}_h$;
- (b) for all $\gamma \in \mathcal{K}_h$ and all b, b' ,

$$\|\nabla_\gamma \Phi_h(\gamma, b) - \nabla_\gamma \Phi_h(\gamma, b')\| \leq M_h \|b - b'\|_{\text{Desc}(h)},$$

where the descendant norm is the maximum of the coordinate norms.

For terminal histories set $M_h = 0$.

Lemma 15 (Lipschitz best responses). *Under Assumption 14,*

$$\|\text{BR}_h(b) - \text{BR}_h(b')\| \leq \frac{M_h}{\sigma_h} \|b - b'\|_{\text{Desc}(h)}.$$

Proof. Let $u = \text{BR}_h(b)$ and $v = \text{BR}_h(b')$. The variational inequalities for constrained optimality give

$$\langle \nabla_\gamma \Phi_h(u, b), v - u \rangle \geq 0, \quad \langle \nabla_\gamma \Phi_h(v, b'), u - v \rangle \geq 0.$$

Consequently,

$$\langle \nabla_\gamma \Phi_h(u, b) - \nabla_\gamma \Phi_h(v, b'), u - v \rangle \leq 0.$$

Strong convexity and insertion of $\nabla_\gamma \Phi_h(v, b')$ yield

$$\begin{aligned} \sigma_h \|u - v\|^2 &\leq \langle \nabla_\gamma \Phi_h(u, b) - \nabla_\gamma \Phi_h(v, b), u - v \rangle \\ &\leq \langle \nabla_\gamma \Phi_h(v, b') - \nabla_\gamma \Phi_h(v, b), u - v \rangle \\ &\leq M_h \|b - b'\|_{\text{Desc}(h)} \|u - v\|. \end{aligned}$$

Divide by $\|u - v\|$ unless it is zero. $\square$

Suppose a Jacobi update satisfies the distance error bound

$$\|a_h^{n+1} - \text{BR}_h(a_{\text{Desc}(h)}^n)\| \leq \eta_{h,n}, \quad \bar{\eta}_n := \max_h \eta_{h,n} \rightarrow 0. \quad (13)$$

Define

$$E_t^n := \max_{h \in \mathcal{H}_t} \|a_h^n - a_h^\sharp\|, \quad \Lambda_t := \max_{h \in \mathcal{H}_t} \frac{M_h}{\sigma_h}.$$

Then Lemma 15 gives the explicit triangular recursion

$$E_t^{n+1} \leq \bar{\eta}_n + \Lambda_t \max_{s>t} E_s^n. \quad (14)$$

No condition $\Lambda_t < 1$ is required, because the dependency graph is acyclic. Equation (14) also gives deterministic error propagation for a single inexact backward sweep.

If a local solver certifies an objective gap $\delta_{h,n}$, strong convexity and the optimality variational inequality imply

$$\|a_h^{n+1} - \text{BR}_h(a_{\text{Desc}(h)}^n)\| \leq \sqrt{\frac{2\delta_{h,n}}{\sigma_h}}. \quad (15)$$

For completeness, the following projected-gradient statement corrects the iteration bound in the original note.

Proposition 16 (Projected-gradient local solver). *Fix h and b . Suppose $\phi(\gamma) := \Phi_h(\gamma, b)$ is twice continuously differentiable on a convex neighborhood of $\mathcal{K}_h$ and*

$$\sigma_h I \preceq \nabla^2 \phi(\gamma) \preceq L_h I \quad \text{for all } \gamma \in \mathcal{K}_h,$$

with $0 < \sigma_h \leq L_h$. Let

$$u^{k+1} = P_{\mathcal{K}_h} \left(u^k - \frac{2}{L_h + \sigma_h} \nabla \phi(u^k) \right), \quad q_h := \frac{L_h - \sigma_h}{L_h + \sigma_h}.$$

Then, with $u^* = \text{BR}_h(b)$,

$$\|u^k - u^*\| \leq q_h^k \|u^0 - u^*\|.$$

If D_h is the diameter of $\mathcal{K}_h$ and $0 < q_h < 1$, it is enough to take

$$k \geq \max \left\{ 0, \left\lceil \frac{\log(D_h/\eta)}{-\log q_h} \right\rceil \right\}$$

to obtain distance error at most η . If $L_h = \sigma_h$, one projected-gradient step is sufficient.

Proof. The constrained first-order condition is equivalent to

$$u^* = P_{\mathcal{K}_h}(u^* - \alpha \nabla \phi(u^*))$$

for every $\alpha > 0$. By the fundamental theorem of calculus,

$$\nabla \phi(u) - \nabla \phi(u^*) = H(u - u^*), \quad H := \int_0^1 \nabla^2 \phi(u^* + s(u - u^*)) ds.$$

Every eigenvalue of H lies in $[\sigma_h, L_h]$. With $\alpha = 2/(L_h + \sigma_h)$,

$$\|I - \alpha H\|_{\text{op}} \leq \frac{L_h - \sigma_h}{L_h + \sigma_h} = q_h.$$

Nonexpansiveness of Euclidean projection therefore gives

$$\|u^{k+1} - u^*\| \leq q_h \|u^k - u^*\|.$$

Iteration proves the bound. The complexity statement follows from $\|u^0 - u^*\| \leq D_h$; when $q_h = 0$, the contraction sends every point to u^* in one step. $\square$

7 f -divergence regularization

We now give a finite class for which the computed all-histories equilibrium is automatically the unique BH equilibrium. Fix a history h and write the next joint state as z . Let ω denote the remaining tail. For a descendant policy b , write

$$P_{\gamma,b}(z, \omega) = \gamma_z q_b(\omega \mid z).$$

Let the product-reference continuation law be

$$Q_h(z, \omega) = m_h(z) R_h(\omega \mid z) > 0, \quad m_h(z) = m_h^X(x) m_h^Y(y).$$

Consider

$$\Phi_h(\gamma, b) = C_h(\gamma, b) + \lambda \sum_{z, \omega} Q_h(z, \omega) f\left(\frac{\gamma_z q_b(\omega \mid z)}{Q_h(z, \omega)}\right), \quad \lambda > 0. \quad (16)$$

For the regularized transport objective of [1, Section 2.1.1], C_h is the expected continuation cost and is linear in γ for fixed b .

Proposition 17 (Unique and interior local minimizers). *Assume for every h :*

- (a) C_h is jointly continuous, convex in γ , and uniformly Lipschitz in γ ;
- (b) $f : [0, \infty) \rightarrow \mathbb{R}$ is continuous and strictly convex;
- (c) f is differentiable on $(0, \infty)$ and $f'(0+) = -\infty$.

Then the objective in (16) is jointly continuous, has a unique minimizer in γ for every b , and that minimizer lies in $\text{ri } \mathcal{K}_h$.

Proof. The map $(\gamma, b) \mapsto P_{\gamma,b}$ is polynomial in the finitely many policy entries. Since Q_h is strictly positive and f is continuous at zero, the divergence term is jointly continuous.

Fix b . If $\gamma \neq \gamma'$, choose a cell z with $\gamma_z \neq \gamma'_z$. Because $q_b(\cdot \mid z)$ is a probability distribution, there is an ω with $q_b(\omega \mid z) > 0$. Strict convexity of f then makes the corresponding summand strictly convex as a function of γ_z . Thus the divergence term, and hence $\Phi_h(\cdot, b)$, is strictly convex. Compactness of $\mathcal{K}_h$ gives a unique minimizer.

Let γ^* be that minimizer and suppose $\gamma_z^* = 0$ for some active cell z . Choose $\bar{\gamma} \in \text{ri } \mathcal{K}_h$, for example the independent coupling $\bar{\gamma}(x, y) = m_h^X(x) m_h^Y(y)$, and set

$$\gamma^\varepsilon = (1 - \varepsilon) \gamma^* + \varepsilon \bar{\gamma}.$$

For every zero coordinate z and every ω with $q_b(\omega \mid z) > 0$,

$$\frac{Q_h(z, \omega) [f(\varepsilon \bar{\gamma}_z q_b(\omega \mid z) / Q_h(z, \omega)) - f(0)]}{\varepsilon} \rightarrow -\infty$$

as $\varepsilon \downarrow 0$, because the right derivative of f at zero is $-\infty$. Contributions from positive coordinates have finite directional derivatives, and the Lipschitz term C_h changes by at most $O(\varepsilon)$. Hence

$$\frac{\Phi_h(\gamma^\varepsilon, b) - \Phi_h(\gamma^*, b)}{\varepsilon} \rightarrow -\infty,$$

contradicting optimality for all sufficiently small $\varepsilon > 0$. Therefore $\gamma^* \in \text{ri } \mathcal{K}_h$. $\square$

Corollary 18 (A genuine finite BH convergence theorem). *Under the hypotheses of Proposition 17 at every history, the finite regularized problem has a unique BH equilibrium policy. Inexact backward sweeps and the parallel Jacobi iteration converge to it whenever their local objective gaps tend to zero. The induced plans converge in total variation.*

Proof. Proposition 17 gives Assumptions 6 and 12. Apply Theorems 8, 9, and 13, followed by Lemma 10. $\square$

The barrier condition applies to several standard generators:

$$\begin{aligned} \text{Kullback–Leibler:} \quad & f(u) = u \log u, & f'(0+) &= -\infty, \\ \text{squared Hellinger:} \quad & f(u) = (\sqrt{u} - 1)^2, & f'(0+) &= -\infty, \\ \text{Jensen–Shannon:} \quad & f(u) = u \log \frac{2u}{1+u} + \log \frac{2}{1+u}, & f'(0+) &= -\infty. \end{aligned}$$

The squared Hellinger and Jensen–Shannon objectives are among the genuinely time-inconsistent examples in [1]. By contrast, the quadratic generator $f(u) = \frac{1}{2}(u - 1)^2$ and the Le Cam generator have finite right derivative at zero. They may still give unique local minimizers and convergence to the all-histories equilibrium, but Proposition 17 does not imply BH uniqueness for them.

For smooth generators, the Hessian calculation in the original note is valid after its domain is stated correctly. At an interior point,

$$\frac{\partial^2}{\partial \gamma_z^2} D_f(P_{\gamma,b} \| Q_h) = \sum_{\omega} \frac{q_b(\omega | z)^2}{Q_h(z, \omega)} f''\left(\frac{\gamma_z q_b(\omega | z)}{Q_h(z, \omega)}\right). \quad (17)$$

If $f'' \geq m_f > 0$ on the attainable ratio range, then Cauchy–Schwarz gives

$$\frac{\partial^2}{\partial \gamma_z^2} D_f(P_{\gamma,b} \| Q_h) \geq \frac{m_f}{m_h(z)} \geq m_f. \quad (18)$$

If in addition $f \in C^2([0, U])$ with bounded f'' , and the remaining terms are C^2 with bounded Hessians, then the quantitative assumptions of Section 6 hold with finite instance-dependent constants.

Remark 19 (Why entropy was not covered by the original projected-gradient claim). For $f(u) = u \log u$, one has $f''(u) = 1/u$, which is unbounded at zero, and the Euclidean gradient is not defined on boundary points. The same global smoothness issue occurs for squared Hellinger and Jensen–Shannon generators. Thus Proposition 16 cannot be invoked on the full transportation polytope without an additional interior argument and an appropriate interior-preserving solver. Corollary 18 avoids this error: the outer convergence theorem is solver-agnostic and requires only certified local objective gaps tending to zero. Mirror, proximal, or interior-point methods may be used locally, but their complexity analysis is separate.

8 Scope and conclusion

The corrected conclusions are as follows.

- (1) The triangular best-response argument is valid for the all-histories policy equations. Approximate local global minimization is stable, and no global contraction is required.
- (2) The all-histories limit always induces a BH equilibrium, so the method computes an actual equilibrium plan in the authors’ sense.

- (3) Without interiority, the BH equilibrium need not be unique, even under all the smooth strong-convexity assumptions used in the original note. Proposition 4 is an explicit counterexample.
- (4) Under interior best responses, the two equilibrium notions coincide. Barrier-type finite f -divergence problems in Proposition 17 satisfy this condition and therefore admit a genuine convergence theorem to the unique BH equilibrium.
- (5) The result remains partial progress. It does not resolve convergence in the semi-discrete or continuous settings of [1], does not analyze convergence under discretization, and does not handle arbitrary nonunique or nonconvex local objectives.

References

- [1] E. Bayraktar and B. Han. Equilibrium transport with time-inconsistent costs. *Mathematics of Operations Research*, 2025. DOI: [10.1287/moor.2023.0323](https://doi.org/10.1287/moor.2023.0323); arXiv: [2302.01498](https://arxiv.org/abs/2302.01498), version 6.
- [2] Open Problems in Operations Research. Problem W4411449955_p0: convergence guarantees for numerical algorithms for time-inconsistent equilibrium transport problems. [Problem page](#).